4. RESULTS

This section reports experimental findings in the order established in Section 3.9. E1 first establishes that Adam-NV performs competitively against existing first-order adaptive optimizers under unperturbed training, prior to any claim regarding its adaptive mechanism. E2 then isolates the mechanism itself, first confirming that Adam-NV's controller responds to injected gradient noise as designed (H2a), and second testing whether that response translates into improved final accuracy relative to standard Adam (H2b). E7-A and E7-C examine the role of time-varying adaptation directly, through a fixed-versus-adaptive ablation and a continuous sweep of adaptation strength. E9 quantifies the computational cost of the mechanism. E11 and E12 test whether the same mechanism generalizes to two further, structurally distinct sources of training-time noise: label corruption and batch-size-intrinsic gradient noise. Findings are reported here without interpretation; their implications are discussed jointly in Section 5.

4.1 E1 — Baseline Convergence Benchmarking

Question. Is Adam-NV's behavior on unperturbed CIFAR-10 training competitive with established first-order adaptive optimizers?

Figure 4 shows provisional trailing-five-epoch validation accuracy. The repository contains four Adam-NV runs but only two complete runs for Adam, AdamW, and AdaBelief; inferential E1 comparisons are therefore withheld pending completion of baseline seeds 2 and 3.

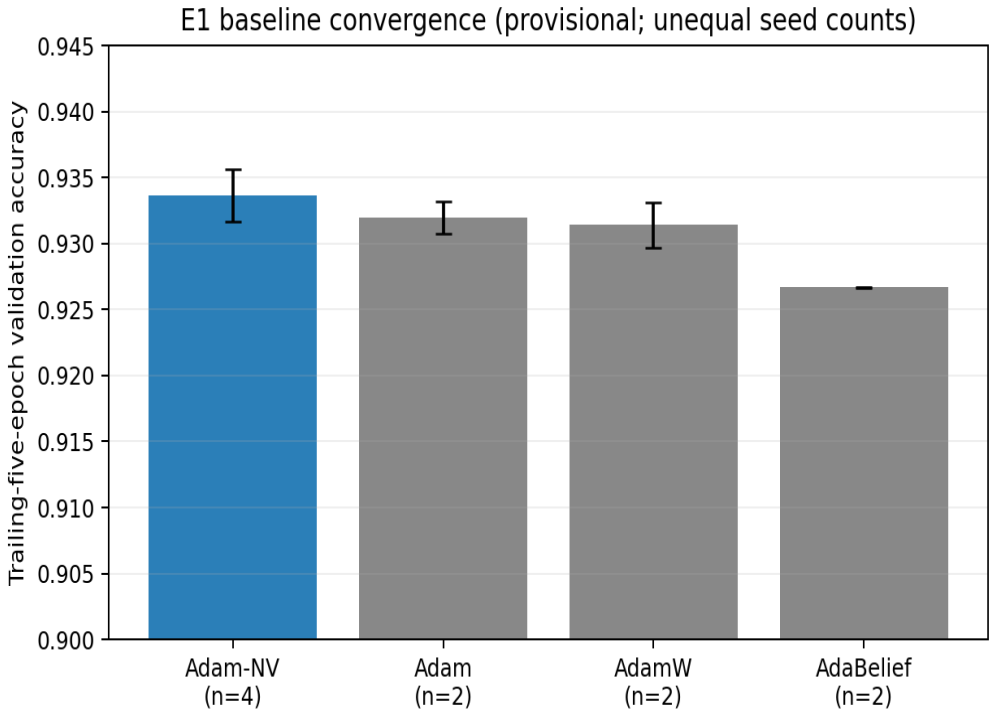


Figure 4. Provisional E1 comparison using available runs; unequal seed counts preclude confirmatory optimizer comparisons.

Table 2. Provisional E1 baseline summary using available repository logs.

Optimizer	Trailing-5 val. acc.	Available n	Confirmatory comparison
Adam-NV (ours)	0.9337 +/- 0.0020	4	Withheld - incomplete baselines
Adam	0.9319 +/- 0.0012	2	Withheld - incomplete baselines
AdamW	0.9314 +/- 0.0017	2	Withheld - incomplete baselines
AdaBelief	0.9267 +/- 0.0000	2	Withheld - incomplete baselines

Findings. Adam-NV has four complete runs; Adam, AdamW, and AdaBelief each have two. The previously reported n=4 baseline table and significance claims are unsupported by the repository snapshot and are withdrawn. No E1 optimizer-superiority or equivalence claim should be made until seeds 2 and 3 are completed for every baseline and paired analysis is rerun.

4.2 E2 (H2a) — Mechanism Validation Under Gradient Noise

Question. Does Adam-NV's adaptive controller respond to injected gradient noise, and if so, at what noise magnitude?

E7-A directly tests whether time-varying adaptation, specifically, contributes to any behavioral or performance difference of the kind observed here.

Figure 5 shows mean β_1 trajectories across all four tested noise levels over 60 epochs of training; mean σ^2 followed a qualitatively similar pattern across the same conditions (Table 5).

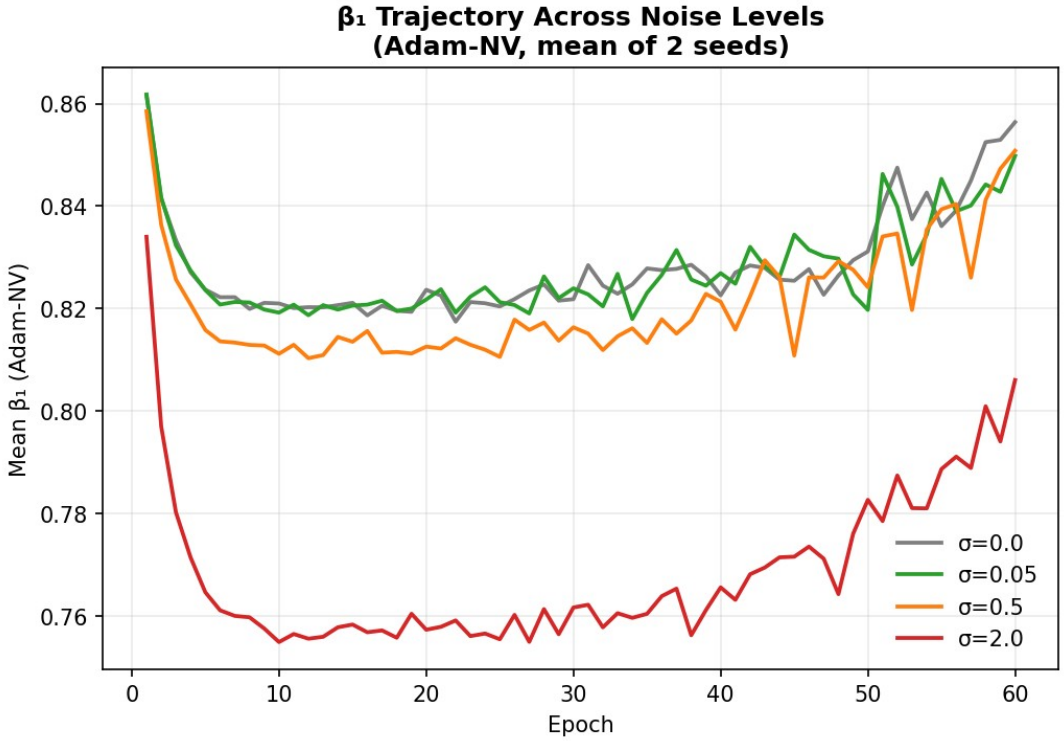


Figure 5. Mean β_1 trajectories across all four tested noise levels over 60 epochs of training (Adam-NV, mean of 2 seeds).

Table 5. Final-epoch mechanism state by noise level.

σ	Final β_1 (approx.)	β_1 / σ^2 response relative to σ = 0.0
0.0	~0.86	—
0.05	~0.85	Not distinguishable
0.5	~0.81–0.84	Modest separation
2.0	~0.72–0.80	Pronounced, sustained separation

Findings. Figure 5 shows β_1 separating by noise level only above a threshold near σ = 0.5; trajectories at σ = 0.0 and σ = 0.05 remain indistinguishable throughout training, while σ = 2.0 produces a pronounced, sustained separation (Table 5). Mean σ^2 followed the same pattern. Per-step diagnostics confirmed injected noise reached the optimizer at the calibrated magnitude at every tested σ.

E2 (H2b) reports whether this mechanism-level response corresponds to a difference in final accuracy between Adam-NV and Adam.

4.3 E2 (H2b) — Performance Robustness Under Gradient Noise

Question. Does Adam-NV's final accuracy degrade less than Adam's under injected gradient noise?

Figure 6 shows validation accuracy curves for Adam and Adam-NV across all four noise levels; Figure 6b shows the corresponding training-loss curves for the same conditions.

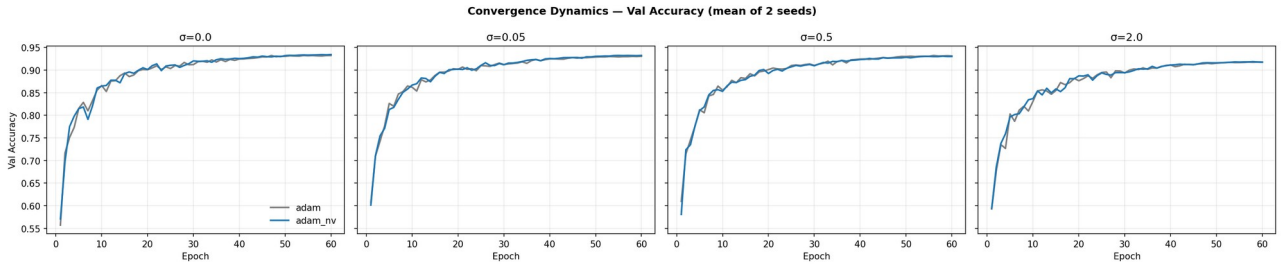


Figure 6. Validation accuracy curves for Adam and Adam-NV across all four noise levels (mean of 2 seeds).

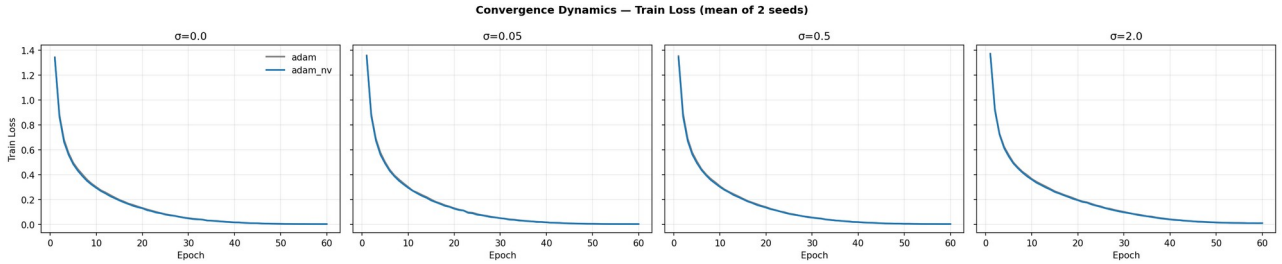


Figure 6b. Training loss curves for Adam and Adam-NV across all four noise levels (mean of 2 seeds), corresponding to the same σ sweep as Figure 6.

Table 6. Δacc comparison, Adam vs. Adam-NV.

σ	Adam val. acc.	Adam-NV val. acc.	Δacc (Adam)	Δacc (Adam-NV)
0.0	0.9324	0.9344	—	—
0.05	0.9306	0.9324	0.0018	0.0017
0.5	0.9313	0.9302	0.0003	0.0038
2.0	0.9174	0.9177	0.0145	0.0158

Findings. Table 6 shows Adam-NV's Δacc was not smaller than Adam's at any tested noise level, including σ = 2.0, where the mechanism response is most pronounced (Figure 5). A power analysis at σ = 2.0 (pooled std ≈ 0.0025) indicates approximately 100 seeds per condition would be needed to resolve a difference of the observed magnitude.

E7-A and E7-C examine whether adaptation strength, specifically, affects outcomes under conditions where the mechanism is confirmed active (σ = 2.0).

4.4 E7-A — Adaptive vs. Fixed β_1 Ablation

Question. Is any behavior of Adam-NV attributable to time-varying adaptation of β_1, as distinct from Adam-NV converging on an effective constant value?

Figure 7 compares final validation accuracy between full adaptive Adam-NV and a fixed-β_1 variant, obtained by setting $\gamma = 0$ so that β_1 remains at its configured constant for the duration of training (Section 3.9), at both tested noise levels.

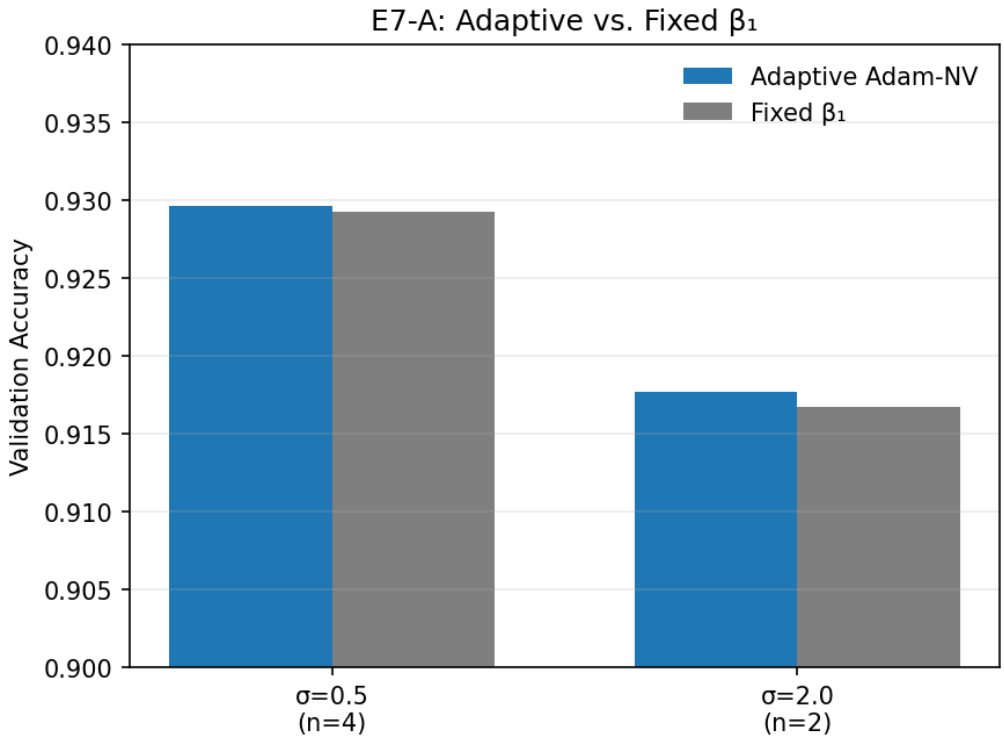


Figure 7. Final validation accuracy, adaptive Adam-NV vs. a fixed-β_1 variant, at both tested noise levels.

Table 7. Adaptive vs. fixed-β_1 comparison.

σ	Fixed β_1	Δacc (adaptive − fixed)	Epochs to 90% (adaptive / fixed)	Cross-seed std (adaptive / fixed)
0.5	0.8366	0.0004	18, 19, 20, 23 / 18, 18, 22, 22	0.0021 / 0.0017
2.0	0.79	0.0009	32, 33 / 32, 33	0.0025 / 0.0017

Findings. Figure 7 and Table 7 show no resolvable advantage for the adaptive configuration over the fixed-β_1 configuration in final accuracy, convergence speed, or cross-seed stability at either noise level. Per-step logging confirmed the fixed variant held β_1 at the configured constant throughout training.

E7-C extends this comparison across a continuous range of adaptation strength (γ) rather than a single on/off contrast.

4.5 E7-C — Adaptation-Strength (γ) Sweep

Question. How do mechanism state and accuracy vary with adaptation strength γ?

Figure 8a and Figure 8b show mean β_1 and mean σ^2 trajectories for $\gamma \in \{0, 0.5, 1, 2\}$; Figure 8c and Figure 8d show the corresponding accuracy trend.

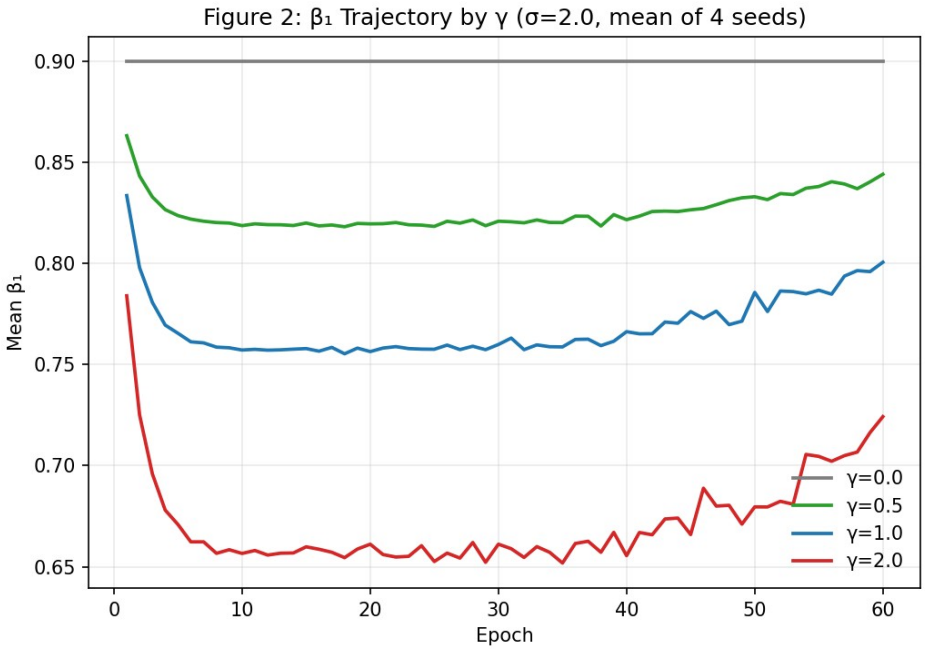


Figure 8a. Mean β_1 trajectory by γ (σ = 2.0, mean of 4 seeds).

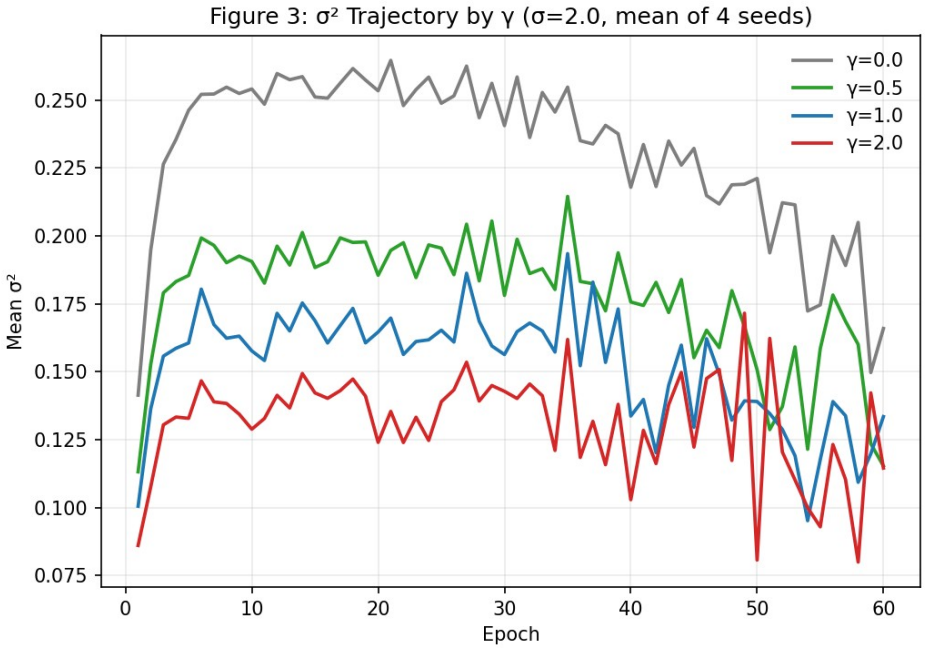


Figure 8b. Mean σ^2 trajectory by γ (σ = 2.0, mean of 4 seeds).

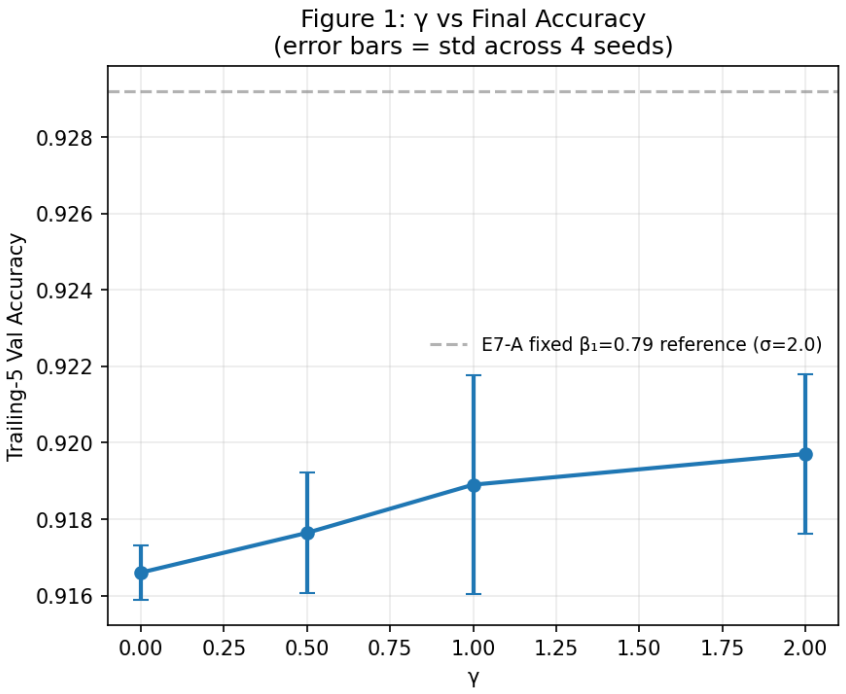


Figure 8c. Trailing-5 validation accuracy as a function of γ (error bars: standard deviation across 4 seeds; dashed line: E7-A fixed-β_1 reference at σ = 2.0).

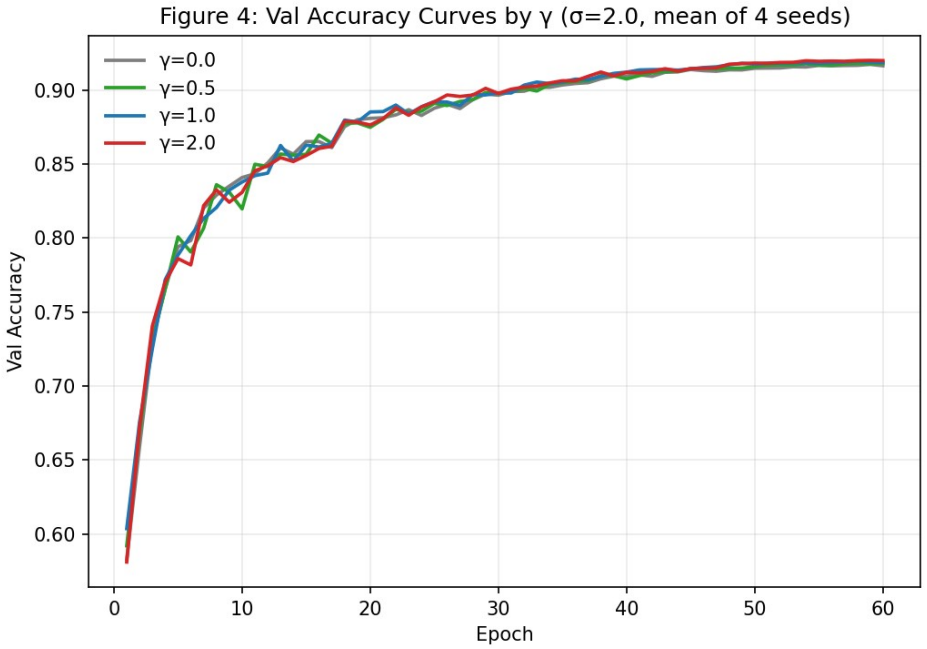


Figure 8d. Validation accuracy curves by γ (σ = 2.0, mean of 4 seeds).

Table 9. E7-C mechanism and accuracy by γ (n = 4).

γ	Final β_1	Final σ^2	Trailing-5 val. acc.
0.0	0.9000	0.1660	0.9166
0.5	0.8442	0.1154	0.9176
1.0	0.8007	0.1335	0.9189

γ	Final β_1	Final σ^2	Trailing-5 val. acc.
2.0	0.7242	0.1145	0.9197

Findings. Figure 8a and Figure 8b show mean β_1 and mean σ^2 varying with γ — monotonically for β_1, non-monotonically for σ^2 (Table 9). Figure 8c and Figure 8d show final accuracy trending upward with γ. The pre-specified comparison (γ = 0 vs. γ = 1) showed no resolvable difference (t = 1.558, p = 0.207). A post-hoc comparison at γ = 2.0, the empirically highest-accuracy value, was likewise not resolvable at n = 4 or n = 8 (p = 0.0535 and p = 0.0865, respectively).

E9 reports the computational cost associated with Adam-NV's mechanism, independent of its effect on accuracy.

4.6 E9 — Computational Overhead

Question. What is Adam-NV's per-epoch computational overhead relative to Adam, and is it constant across training conditions?

Figure 9 shows the overhead ratio (Adam-NV time / Adam time) as a function of batch size, with a horizontal reference line at 1× indicating no overhead.

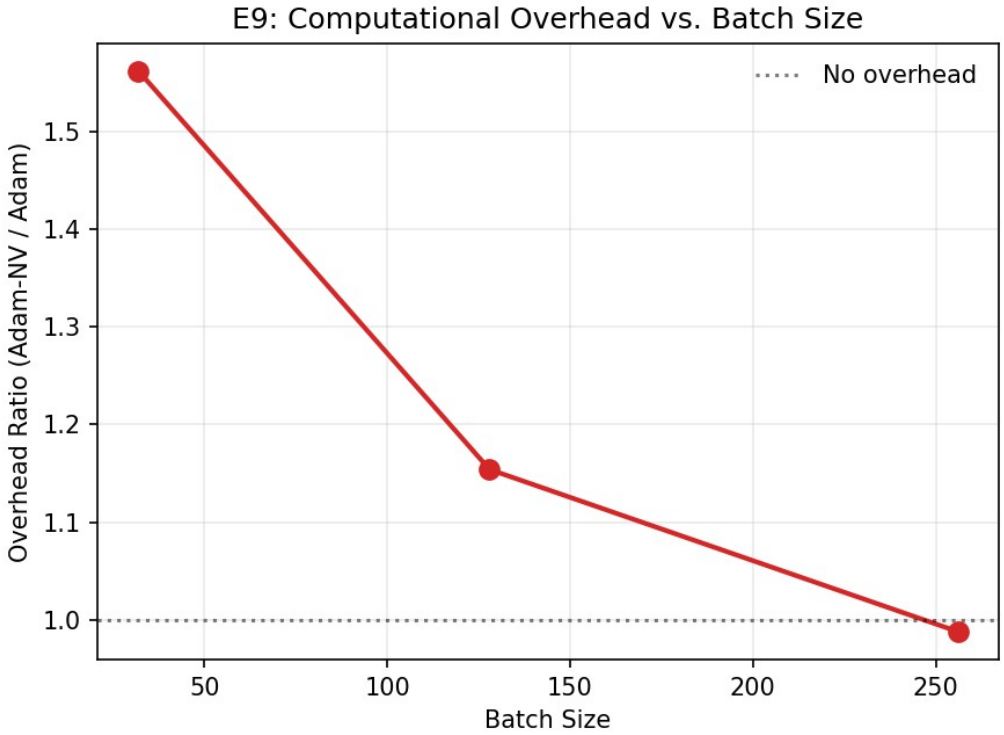


Figure 9. Overhead ratio (Adam-NV time / Adam time) as a function of batch size, with a 1× reference line indicating no overhead.

Table 11. Overhead ratio (Adam-NV / Adam) across tested conditions.

Condition	Adam (s/epoch)	Adam-NV (s/epoch)	Ratio
Batch size 32	50.19	78.38	1.562×
Batch size 128	42.21	48.70	1.154×
Batch size 256	46.48	45.92	0.988×
σ sweep (E2, mean across 4 levels, bs = 128)	—	—	1.21×
Label noise, clean (E11, bs = 128)	—	—	1.18×
Label noise, 20% corrupted (E11, bs = 128)	—	—	1.14×

Findings. The displayed epoch-time ratios are exploratory and vary from 0.988x to 1.562x across batch sizes. Because they were not obtained from repeated synchronized benchmark runs and lack uncertainty estimates, they do not

support a stable-overhead claim. Confirmatory timing remains pending under the repeated warm-up and profiling protocol in Section 3.9.

4.7 E11 — Label-Noise Evaluation

Question. Does Adam-NV's mechanism respond to label corruption, a structurally different perturbation from injected Gaussian gradient noise, and does any such response correspond to improved robustness?

E11 and E12 report mechanism and accuracy behavior under label corruption and batch-size variation respectively, the two remaining perturbation types examined in this study.

Figure 10 shows mean beta_1 and heuristic-score trajectories for clean and 20%-label-corrupted training. The curves cross near epoch 26, but these controller trajectories alone cannot identify a memorization transition.

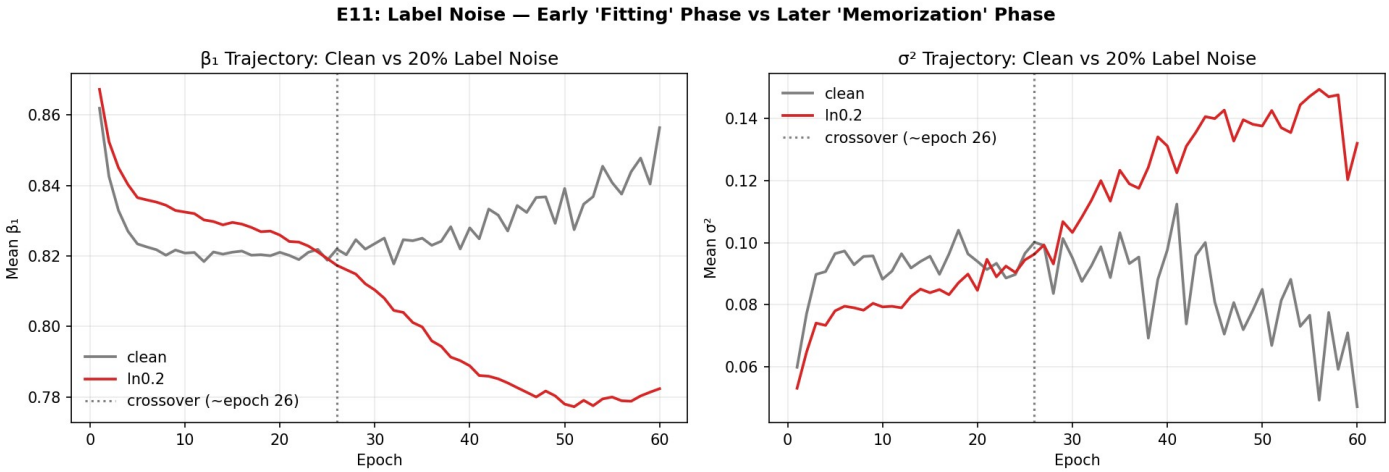


Figure 10. Mean β_1 and mean σ^2 trajectories, clean vs. 20%-label-corrupted training, over 60 epochs.

Table 12. E11 accuracy and mechanism state, clean vs. 20% label-corrupted training.

Metric	Clean	20% Label Noise	Statistic
Adam val. acc.	0.9329 ± 0.0014	0.7863 ± 0.0024	Degradation = 0.1466
Adam-NV val. acc.	0.9323 ± 0.0031	0.7871 ± 0.0064	Degradation = 0.1453
Mean β_1 (final)	0.8452	0.7803	t = 28.18, p = 0.00008
Mean σ^2 (final)	0.0609	0.1391	t = −12.50, p = 0.00002

Findings. Mean beta_1 and the heuristic score diverge between clean and label-corrupted training by epoch 60. The crossover near epoch 26 is a mechanism-state observation, not direct evidence of memorization. The claim remains provisional until training accuracy is measured separately on uncorrupted examples and corrupted examples against both clean and reassigned labels. No detectable robustness advantage was found in the reported interaction analysis.

E12 examines whether the same mechanism responds to gradient noise arising intrinsically from minibatch sampling, rather than from an explicit perturbation (gradient noise, in E2; label corruption, here).

4.8 E12 — Batch-Size Sensitivity

Question. Does Adam-NV's controller respond to gradient noise that arises intrinsically from minibatch sampling, and does batch size affect accuracy or computational overhead?

Figure 11 shows mean β_1 and mean σ^2 as a function of batch size.

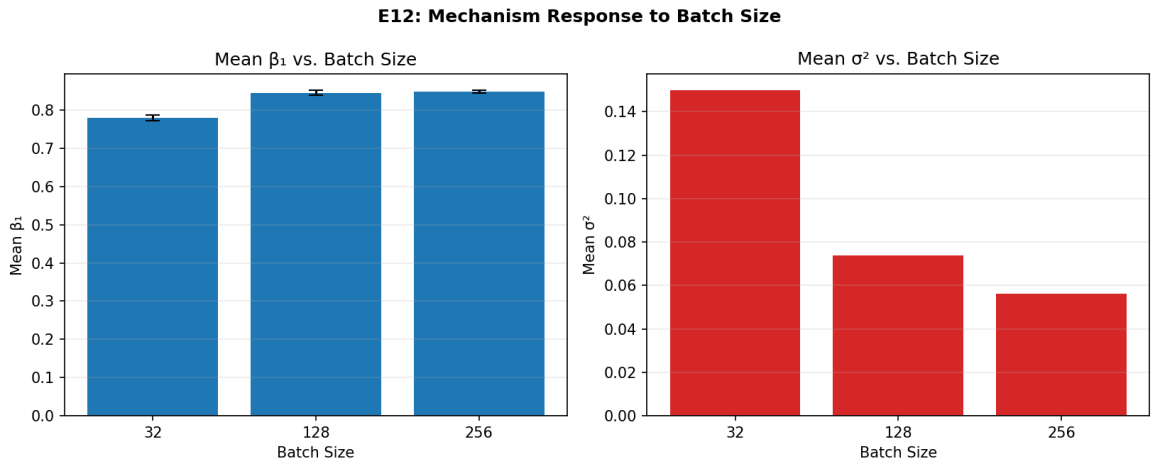


Figure 11. Mean β_1 and mean σ^2 as a function of batch size.

Table 13. Mechanism and accuracy by batch size (n = 4).

Batch size	Mean β_1 (± std)	Mean σ^2	Δacc (Adam-NV − Adam)	p (Δacc)
32	0.7802 ± 0.0073	0.1499	0.0004	0.65
128	0.8454 ± 0.0057	0.0738	0.0005	0.72
256	0.8482 ± 0.0039	0.0563	0.0006	0.50

Findings. Figure 11 and Table 13 show mean β_1 differing between batch size 32 and batch sizes 128 and 256 ($p \leq 0.00003$ for both comparisons), consistent with a possible threshold effect rather than a continuously graded response. β_1 did not differ resolvably between batch sizes 128 and 256 ($p = 0.45$), despite a twofold difference in per-epoch step count. No resolvable Δacc between Adam and Adam-NV was observed at any batch size tested.

This concludes the experimental results. Table 14 summarizes mechanism-validation and performance outcomes across the noise-response experiments reported above; Section 5 discusses these findings jointly, including their cross-experiment consistency and the limitations of the present study.

Table 14. Summary of mechanism validation and performance outcomes across noise-response experiments.

Experiment	Mechanism Response Confirmed	Resolvable Accuracy Improvement over Adam
E2	✓	✗
E7-A	✓	✗
E7-C	✓	✗
E11	✓	✗
E12	✓	✗

4.9 E13-E16 - Extended Study Status

E13 non-stationary schedules, E14 coupled beta_2 sensitivity, and E15 decoupled noise memory are present in the repository and have been reanalyzed with paired seed matching. In the current exploratory table, no E13 Adam-NV-versus-Adam contrast and no E15 decoupled-versus-coupled contrast survives Holm correction. E14 suggests a ramp-up degradation at beta_2=0.95 in the adjusted paired t-test, but the exact four-seed sign-flip p-value is 0.125. One intermittent E14 run ends at epoch 32, so the analysis correctly refuses publication-mode output. E16 CIFAR-100 and all protocol-v2 lag reruns remain pending. These extended results must replace, not be combined with, the invalid legacy 1,317-to-47-step lag claim.

5. Discussion

5.1 Synthesis of Findings

Across E2, E7-A, E7-C, E11, and E12, Adam-NV's logged controller state changed with the tested perturbations. These results validate activation of the implemented mechanism, not calibration of sigma^2 as variance or optimality of lowering beta_1 under noise. No tested condition established a detectable validation-accuracy advantage over Adam.

5.2 Practical Implications

The current evidence does not support Adam-NV as a general drop-in replacement for Adam for improved noisy-training accuracy. Earlier overhead values are exploratory; formal synchronized benchmarking is pending. Adam-NV is best viewed at present as a mechanism probe whose controller direction, generality, and runtime cost still require validation.

5.3 Why Mechanism Does Not Imply Performance

A plausible explanation is that Adam's existing per-parameter second-moment normalization already absorbs much of the effect that β_1 adaptation targets. If the denominator term already down-weights unreliable directions, an additional, first-moment-level noise response may be largely redundant rather than additive. This is consistent with the absence of a resolvable accuracy difference even at $\sigma = 2.0$, where the mechanism response is most pronounced (E2, E7-A).

5.4 Why β_1 Adaptation Alone May Be Insufficient

Noise robustness in adaptive optimizers is typically distributed across several interacting terms — the first-moment decay rate, the second-moment decay rate, the effective learning rate, and any gradient clipping applied. Adapting only β_1 leaves these other terms fixed, which may limit the mechanism's ability to influence final accuracy on its own. A controller that jointly adapts β_1 and β_2, or that couples β_1 adaptation to the learning-rate schedule, may be a necessary extension rather than an incremental one.

5.5 Why Optimizer Competitiveness Does Not Imply Mechanism Benefit

The available E1 repository logs are incomplete for the baseline optimizers, so competitiveness claims are provisional. Across the completed noise-response experiments, no detectable accuracy advantage was found under the tested conditions.

5.6 Future Controller Designs

Extensions worth testing directly from these results include a joint β_1–β_2 controller, per-layer rather than per-tensor adaptation strength, and a controller tuned specifically to respond to label corruption rather than to zero-mean gradient noise, given the qualitatively different trajectory shape observed in E11. Coupling the noise estimate to an adaptive learning-rate reduction, rather than to β_1 alone, is also a natural next step given the discussion in Section 5.4.

5.7 Threats to Validity

Several factors limit the generality of these conclusions:

- **Architecture.** All experiments use a single CIFAR-adapted ResNet-18; results may not transfer to other architectures (e.g., transformers, wider or deeper networks).
- **Dataset.** CIFAR-10 is the sole primary dataset; label noise and gradient noise were studied on this dataset alone.
- **Batch-size / step-count coupling.** E12's batch-size conditions co-vary with per-epoch step count at a fixed epoch budget (Section 3.9), so batch-size and step-count effects cannot be fully separated.
- **Optimizer family.** Comparisons are limited to Adam-family optimizers (Adam, AdamW, AdaBelief); no comparison was made to non-adaptive methods (e.g., SGD with momentum).
- **Epoch budget.** All experiments use a fixed 60-epoch budget; mechanism or performance differences that emerge only over longer training schedules would not be captured here.

Priority follow-up work is to complete missing baseline and E14 runs, rerun the corrected protocol-v2 response diagnostic across seeds, test the reverse controller direction, collect direct label-memorization metrics, benchmark overhead formally, and complete CIFAR-100 evaluation. E13-E15 statistical conclusions must use paired confidence intervals, direct contrasts, and Holm-adjusted p-values.

6. Conclusion

This study introduced Adam-NV, which varies beta_1 per parameter element and step using a bounded heuristic noise score. The original rule lowers beta_1 as the score rises. The score is not a calibrated variance or noise-to-signal estimator, and the optimal direction of the controller remains an empirical question.

The reported CIFAR-10 experiments show that the controller state responds to several perturbation types. They do not establish optimizer superiority. E1 baseline comparisons remain provisional because only Adam-NV has four complete runs in the repository snapshot.

No detectable improvement in final validation accuracy over Adam was found under the tested conditions and available seed counts. This wording does not imply equivalence or rule out smaller effects.